ISSSP 2026 — TRACK 3 FINALS: ITEMS 3 & 4 (TOPIC SELECTION & RESEARCH CONTENT — 40 POINTS)

L-MINER: Autonomous Multi-Tier Rover for Subsurface Lunar Volatile Prospecting and Mineral Classification

Technical Research Paper & Engineering Program: Dual-Modality Radar-Spectrometer Sounding, Edge Machine Learning, and Hierarchical Autonomy for In-Situ Resource Utilization (ISRU)

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Scoring Focus: Items 3 (10 Pts) & 4 (30 Pts) — Total 40 Pts	**Target Environment:** Lunar South Pole Cold Traps (<40 K)

ABSTRACT — In-situ prospecting of volatile water ice in lunar South Pole Permanently Shadowed Regions (PSRs) is critical for sustainable space exploration. However, extreme cold (<40 K), lack of solar illumination, and teleoperation latency prohibit manual operations. This paper presents **L-MINER Mk-II**, an autonomous multi-tier rover fusing 500 MHz Ground Penetrating Radar (0–3 m) and active 0.5–2.5 μm VNIR-SWIR reflectance spectrometry. A 3-tier decoupled autonomy framework coordinates L0 collision reflexes (10 Hz), L1 stochastic power/thermal management (1 Hz), and L2 cognitive science sounding. Onboard signal processing compresses 32,968 raw floats into a 10-D feature vector (3,300x compression), classified via Random Forest with 95.8% accuracy in <15 ms. Across a 60 x 60 m surveyed grid, the system identifies ~57.8 Tonnes of recoverable water ice (yielding ~51.4 Tonnes LOX + ~6.4 Tonnes LH2 propellant), providing a validated roadmap for autonomous lunar prospecting.

Part I: Topic Selection Justification & Mission Significance (10 Points)

1.1 The Ground-Truth Gap: Orbital spectrometry (M^3) samples only the uppermost 10–50 microns of space-weathered dust, while orbital SAR suffers from ambiguous roughness scattering and zero real-time excavation capability. Commercial and scientific ISRU extraction requires ground-truth vertical stratification (overburden thickness and ice purity) achievable solely through surface mobile rovers.
1.2 Autonomous Multi-Tier Necessity: With a 3-second roundtrip speed-of-light delay and complete communication shadows inside deep polar craters, teleoperation is unfeasible. Local intelligence must autonomously safeguard rover mobility, regulate battery and thermal limits, and execute science.
1.3 Space Roadmap Alignment: Aligns directly with NASA Artemis III, CNSA ILRS/Chang'e-7, and ISRO Chandrayaan-4 polar survey priorities.

Part II: Research Content & Engineering Program (30 Points)

2.1 Hierarchical Decoupled Autonomy: L0 Reflex (10 Hz; HazCam point-cloud hazard scoring H >= 0.75 triggers immediate halt and reroute); L1 Resource Manager (1 Hz; battery SoC < 30% or T > 50°C pauses operations for solar recharge); L2 Science Sounding (On Demand; triggers GPR/Spectrometer, feature extraction, and ML classification).

Layer	Scope	Freq	Decision Threshold	Hardware Subsystem
L0: Reflex	Hazard Bypass	10 Hz	H >= 0.75 or slope > 15° triggers halt	HazCam, Rocker-Bogie
L1: Resource	Power & Thermal	1 Hz	SoC < 30% or T > 50°C pauses for recharge	LiFePO4, Solar Arrays
L2: Science	Sounding & ML	Demand	GPR + Spectrometer 10-D classification	500MHz GPR, VNIR-SWIR

2.2 Sensor Modeling & Synthetic Physics:
• **500 MHz GPR:** Fresnel dielectric reflection $\Gamma = (\sqrt{\varepsilon 1} - \sqrt{\varepsilon 2})/(\sqrt{\varepsilon 1} + \sqrt{\varepsilon 2})$. Water Ice permafrost produces sharp contrast (Γ=0.85, width 3); Pyroxene produces bedrock reflection (Γ=0.65); conductive Ilmenite causes attenuation (Γ=0.55); porous Regolith creates diffuse scattering (Γ=0.30). Generates 128x256 B-scan.
• **VNIR-SWIR Spectrometer (0.5–2.5 μm):** Calibrated to USGS splib07 and Apollo sample returns. Water Ice exhibits strong vibrational O-H absorption dips at 1.25, 1.50, and 2.00 μm; Ilmenite shows flat dark Fe-Ti charge transfer (R ~ 0.07); Pyroxene shows Fe^{2}■ crystal field bands at 1.0 and 2.0 μm; space-weathered Regolith displays nanophase iron red slope.
2.3 Edge Feature Extraction (3,300x Compression): Reduces 32,968 raw floats into 10 scalar features (GPR: peak amplitude, energy, layer contrast, max depth; Spectrometer: mean albedo, R(1.0), R(1.5), R(2.0), spectral slope, 1.5 μm band depth).

Mineral	Prec	Rec	F1	Key Diagnostic Signal
Water Ice	99.1%	99.3%	0.992	1.5 μm Absorption Band Depth & High GPR Amplitude (A=0.85)
Ilmenite	97.2%	95.8%	0.965	Flat Low Albedo (R < 0.10) & Attenuated Radar Return
Pyroxene	94.6%	95.0%	0.948	Dual 1.0 μm (Band I) and 2.0 μm (Band II) Absorption Dips
Regolith	92.3%	92.8%	0.925	Monotonic Red Slope Patina & Diffuse Subsurface Scattering
OVERALL	**95.8%**	**95.8%**	**0.958**	Random Forest (100 Trees) — Inference Latency < 15 ms on ARM SoC

2.5 Quantitative ISRU Yield: In a 6 x 6 grid (60 x 60 m zone), 7 water ice deposits localized at avg 0.88 m depth. M_{water} = 7 cells x (100 m^2 x 1.0 m x 1500 kg/m^3 x 5.5 wt%) = **~57.8 Tonnes pure H■O**. Electrolysis yields **~51.4 Tonnes LOX + ~6.4 Tonnes LH2 propellant**, sufficient to completely refuel two lunar ascent stages.

Part III: Engineering Verification & Future Roadmap

Roadmap: TRL 4 (Current Cyber-Physical Digital Twin) -> TRL 5 (Q1–Q3 2027: Hardware-in-the-Loop cryo-simulant testing with physical 500 MHz GPR and InGaAs spectrometer) -> TRL 6 (2028: Thermal vacuum chamber qualification & porting to NASA Core Flight System on RTG4 FPGA).

Key References: Pieters et al. (Science 2009, Chandrayaan-1 M³); Clark et al. (USGS splib07); Fang et al. (Chang'e-3 LPR); Hamran et al. (Perseverance RIMFAX); McKay et al. (JSC-1 Simulant).